Prakul Hiremath



Researcher & Engineer · Computer Science

Belagavi, Karnataka, India | B.Tech, Computer Science & Engineering, VTU (2023–27)
prakulhiremath@vtu.ac.in | hiremathprakul.aoe@gmail.com
Google Scholar | GitHub | LinkedIn | OpenReview | Medium

RESEARCH PROFILE

I study how intelligent systems work, where they fail, and what is lost when uncertainty is compressed into simplified representations. My research interests connect information theory, geometry, machine learning evaluation, and sequential decision-making. I am interested in understanding when information-preserving assumptions break down and how those failures change model decisions.

RESEARCH INTERESTS

Epistemic uncertainty and calibration; information loss in sequential systems; representation geometry and robustness under distribution shift; evaluation metrics and capability detection; reinforcement learning and model behaviour.

EDUCATION

B.Tech in Computer Science and Engineering 2023–2027

Visvesvaraya Technological University (VTU), Belagavi, Karnataka, India

CGPA: 8.02/10

SELECTED PUBLICATIONS & RESEARCH MANUSCRIPTS

When Does Sequential Detection Collapse to a Scalar? A Necessary and Sufficient Characterisation

NeurIPS 2026 · Accepted poster

Studies conditions under which sequential detection can be represented by a scalar statistic without losing decision-relevant information.

Poster page · Sydney poster session: 10 December 2026, 10:00 AM–1:00 PM AEDT.

Latency-Conditioned Selection Bias in RLHF

NeurIPS 2026 · Accepted poster

Studies how rollout latency and asynchronous data collection can introduce selection bias in reinforcement learning from human feedback.

Poster page · Sydney poster session: 8 December 2026, 5:00–8:00 PM AEDT.

Evaluation Collapse: When Metrics Cannot Distinguish Model Behavior

ICLR 2027 · Submitted

Examines when scalar evaluation metrics fail to distinguish systems with meaningfully different behaviour.

OpenReview · Manuscript PDF

Information Is Not Enough: A Tight, Post-Hoc Diagnostic for Representation Robustness Under Distribution Shift

AISTATS 2027 · Submitted

A post-hoc diagnostic for assessing whether frozen representations preserve task-relevant information under anticipated distribution shifts.

OpenReview · Manuscript PDF

Calibration Drift Under Reasoning: How Chain-of-Thought Budgets Induce Overconfidence in Large Language Models

arXiv · Preprint

arXiv:2606.11211

When Confidence Metrics Fail: Sensitivity–Uncertainty Alignment as a Diagnostic for High-Risk Model Errors

NeurIPS 2026 E&D · Submission

Studies alignment between uncertainty estimates and model sensitivity to input changes in high-risk prediction settings.

arXiv preprint · Code

RESEARCH EXPERIENCE

Capability Dynamics Researcher · ITISMU, VTU Belagavi

Research

Working with Dr. M. N. Birje on how model capabilities emerge during training and how detection timing depends on evaluation resolution. Studies the distinction between a capability first becoming detectable and becoming reliably usable across training checkpoints.

Astroinformatics Researcher · Astronomy and Calculus Club, VTU

Conference presentation

Presented “AI-Driven, Uncertainty-Aware Anomaly Detection in Multi-wavelength AstroSat Observations” at *A Decade of AstroSat Observations: Science Outcomes and Future Prospects*, organised by the U. R. Rao Satellite Centre (URSC), ISRO, Bengaluru, 30 January–1 February 2026.

Conference information

Bioinformatics Research · Aliens on Earth

Applied research

Contributed to BIOLOOP, an AI-enabled algal biorefinery concept for industrial CO_2 capture and biomass production, and to S2N, a seawater-based green-hydrogen initiative. BIOLOOP was included in an AI-in-energy case study for the India AI Impact Summit 2026.

Presented “Real-Time Monitoring of Bioaerosol Emissions in Hydrogen Reactors” at the 14th Asian Aerosol Conference (AAC 2025), Mumbai, December 2025.

Global AI Impact Commons case study · Poster story · Casebook announcement

SELECTED RESEARCH SOFTWARE & PROJECTS

LOB-Latent-Regimes

Financial systems

Research on latent microstructure regimes and early signals in high-frequency limit order book data. Paper · Repository

Temporal Leaks

Python · Time series

Tool for detecting look-ahead bias and accidental use of future data in time-series machine-learning pipelines. PyPI · Repository · Project note

Latency Gym

C++ · Reinforcement learning

A C++ and Gymnasium environment for exploring reinforcement-learning approaches to latency optimisation in trading infrastructure. Repository · Project note

ParallelWatch

Python · Anomaly detection

Multivariate temporal correlation engine for anomaly detection across related infrastructure and financial metrics. PyPI · Repository

Flashback

Python · Data engineering

Time-travel debugging and transformation-lineage tracking for Pandas and Polars workflows. PyPI · Repository

HED Score

Evaluation

A time-aware metric designed to evaluate when a detector identifies an event alongside detection quality. Paper · PyPI · Colab

BIOLOOP

Climate · Biotechnology

Algae-based approach to industrial CO_2 capture and biomass production. Contributed before exiting the initiative.

S2N

Clean energy

Seawater-based green-hydrogen initiative exploring pathways for hydrogen production.

PROJECT FUNDING & RESEARCH AWARDS

₹15 lakh	**MSME Idea Hackathon 4.0 · Indian Institute of Science** S2N seawater-based green-hydrogen project.
₹2.5 lakh	**FKCCI INNO MANTHAN 2026 – 18th Edition** BIOLOOP project grant/prize.
₹1 lakh	**Startup Karnataka NAIN 2.0 · Indian Institute of Information Technology (IIIT) Dharwad** BIOLOOP project.
₹2 lakh	**Startup Karnataka NAIN 2.0 · Indian Institute of Information Technology (IIIT) Dharwad** URJAWAVE project.
₹25,000	**Vice Chancellor's Research Grant · KLE Academy of Higher Education and Research (KAHER)** Project on early detection of gingivitis and dental caries.

PATENTS & REGISTERED DESIGNS

Patent applications

- **202541014688** — Integrating Neuromorphic Computing and Reinforcement Learning in Hybrid AI for Autonomous Systems (2025; application).
- **202541014687** — Innovative CO_2 Emission Filters for Vehicles: Net-Zero Carbon Solutions with Advanced Adsorption and Smart Monitoring (2025; application).
- **202541014690** — Dynamic Adjustable Sleeve D&M for Enhanced Universal CO_2 Emission Filters (2025; application).
- **202541014689** — Hyper-Adaptive Manufacturing Using Multi-Agent AI for Autonomous Production Optimization (2025; application).
- **202541032942** — EMEFF: An Explainable Multimodal Economic Forecasting Framework for Real-Time, Transparent, and Adaptive Predictions (2025; application).
- **202641023750 A** — A Spatial–Temporal Deep Learning Framework for Robust Human Pose Estimation in Crowded Visual Scenes (filed 27 February 2026; published 6 March 2026).

Registered designs

- **463354-001** — Modular Algae Pod Device for Sustainable CO_2 Capture (certificate issued 2 June 2026).
- **463353-001** — Dual-Chamber CO_2 Capture and Air Purification Device Using Algae on Biodegradable Film (certificate issued 25 March 2026).
- **463624-001** — Disposable Algae Bioreactor Device for CO_2 Capture Using Hydrogel and Bioplastics (certificate issued 25 March 2026).

RESEARCH & PROFESSIONAL PROFILES

Google Scholar: scholar.google.com/citations?user=ERbZUdoAAAAJ

GitHub: github.com/prakulhiremath

LinkedIn: linkedin.com/in/prakulhiremath

OpenReview: openreview.net/profile

Medium: medium.com/@prakulhiremath